

Whole Share Constrained Portfolio Optimization

Andrew Hall
Brigham Young University
Provo Utah, USA
amh1124@byu.edu

Abstract—This paper presents a novel approach to mean-variance portfolio optimization by incorporating whole share constraints, addressing a key limitation of traditional Markowitz optimization models. While classical methods assume the ability to purchase fractional shares, this assumption fails in many real-world scenarios, such as portfolios constrained by whole share trading requirements. We propose a two-stage optimization framework that combines quadratic programming (QP) with mixed-integer quadratic programming (MIQP) to achieve optimal whole share allocations. To evaluate the effectiveness of this method, we conduct a series of experiments using price data sampled from the CRSP historical stock dataset. We introduce backlog risk as a measure of excess risk resulting from discrepancies between fractional and whole share weights. Our results highlight the practical applicability of the proposed method, providing an effective solution for portfolio construction under whole share constraints.

Index Terms—Portfolio Optimization, Quadratic Programming, Mixed Integer Quadratic Programming

I. INTRODUCTION

Since the publication of Harry Markowitz’s “Portfolio Selection” in the Journal of Finance, mean-variance portfolio optimization has become a cornerstone of asset allocation in the finance industry. However, Markowitz’s model does not account for several practical constraints encountered in real-world portfolio construction, such as limits on leverage, restrictions on short selling, and the requirement to trade in whole shares. This paper introduces a novel approach to mean-variance portfolio optimization by incorporating whole share constraints, using mixed-integer quadratic programming (MIQP).

II. BACKGROUND

A. The Markowitz Model

In “Portfolio Selection,” Markowitz asserts that investors value expected returns while seeking to minimize risk (return variance), a principle that underpins most portfolio optimization techniques. While this framework provides valuable insights into asset allocation, it falters when applied to real-world scenarios. One of its key assumptions, that investors can purchase fractional shares, is often unrealistic. For large institutional investors, such as multi-billion-dollar hedge funds, the ability to approximate fractional weights with large positions mitigates this issue. Similarly, fractional share trading, popularized by platforms like Robinhood, has made Markowitz’s model more accessible to retail investors. However, for investors who must trade in whole shares, traditional optimization approaches remain inadequate.

B. The Sharpe Ratio

Coined as the reward-to-variability ratio in 1966, William F. Sharpe introduced a new metric for evaluating mutual fund performance. Over time, this metric has become widely known as the “Sharpe Ratio” and is often formulated as:

$$S = \frac{R_p - R_f}{\sigma_p} \quad (1)$$

where R_p is the portfolio’s expected return, R_f is the risk-free rate of return, and σ_p is the standard deviation of the portfolio’s excess return.

This metric provides investors with an intuitive understanding of an asset or portfolio’s performance by capturing its return adjusted for risk. A higher Sharpe Ratio indicates a more favorable risk-adjusted return, making it a key criterion for portfolio evaluation. Consequently, the goal of portfolio optimization can be framed as a Sharpe Ratio maximization problem, where the objective is to construct a portfolio that delivers the highest possible return for a given level of risk. For the sake of simplicity we will assume the risk-free rate of return is zero for all further calculations since it has no direct impact on the optimization problems at hand.

C. CVXPY

For our research, we chose to use the CVXPY Python library to implement all of our optimization methods. CVXPY is an open-source library designed for solving convex optimization problems in Python. It provides a high-level interface for defining and solving optimization problems with a wide range of constraints and objective functions. One of the key advantages of CVXPY is its flexibility and support for a variety of solvers, including OSQP, SCS, and ECOS, making it a powerful tool for both academic research and real-world applications. By leveraging CVXPY, we were able to seamlessly integrate our constraints, such as whole share requirements and budget limits, into the optimization process. This library also ensured that our implementations were concise, reproducible, and easy to adapt as our research evolved.

III. CLASSICAL PORTFOLIO OPTIMIZATION: QUADRATIC PROGRAMMING

A. Procedure

Quadratic programming (QP) is a standard technique for solving mean-variance portfolio optimization problems. While our goal is to maximize the Sharpe Ratio of our portfolio, we

cannot set the objective function as the Sharpe Ratio directly due to the division operator present in the formula. In order to get around this we use Markowitz's original formulation of investor utility as seen in Equation 2. Our optimization method begins with three inputs:

- 1) A vector of forecasted returns,
- 2) A covariance matrix of forecasted asset returns,
- 3) A loss-aversion parameter, denoted here as γ .

The objective function maximizes portfolio return while penalizing portfolio variance, scaled by the γ parameter. Mathematically, the optimization problem is defined as:

$$\max_{\mathbf{w}} \mathbf{w}^T \mathbf{r} - \gamma \mathbf{w}^T \mathbf{V} \mathbf{w} \quad (2)$$

where $\mathbf{w}$ is the vector of portfolio weights, $\mathbf{r}$ is the vector of expected returns, $\mathbf{V}$ is the covariance matrix, and γ represents the investor's loss aversion parameter.

This formulation produces a vector of optimal portfolio weights. If desired, these weights can be normalized to ensure the portfolio is fully invested by scaling the optimal weights until they sum to one.

B. Loss Aversion

The γ parameter serves as a proxy for investor loss aversion, reflecting the principle that investors dislike losses more than they value equivalent gains. As shown in Figure 1, portfolios with higher loss aversion (higher γ) exhibit lower leverage, aligning with the risk-averse behavior of these investors. Conversely, risk-tolerant investors are more likely to lever their portfolios.

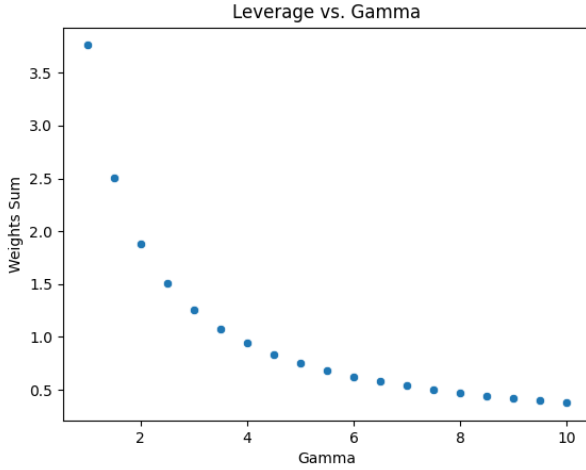


Fig. 1. Illustration of the relationship between loss aversion and leverage.

C. Initial Results

As a preliminary step, we conduct an initial experiment to illustrate the practical challenges of whole share constrained portfolio optimization. Using 10 years of historical pricing data for Apple (AAPL), Ford (F), Coca-Cola (KO), and Verizon (VZ), we derived the expected returns vector and covariance

matrix. First we calculate the QP fractional shares optimal portfolio. Then we apply three different rounding methods: ceiling, floor, and midpoint.

This experiment highlights a critical issue: naive rounding methods, such as rounding up or down to whole shares, can result in portfolios that exceed the budget constraint, which is both impractical and undesirable for real-world implementation.

TABLE I
COMPARISON OF ROUNDING METHODS FOR QUADRATIC PROGRAMMING

Method	Exp. Ret.	Std. Dev.	Value	Deficit
Fractional Shares	23.80	17.79	1000000.00	-0.00
Ceiling	23.80	17.79	1000205.89	205.89
Floor	23.80	17.79	999773.55	-226.45
Midpoint	23.80	17.79	1000062.55	62.55

We also plot the efficient frontier, which represents the set of mean-variance optimal portfolios available to an investor. This curve highlights the trade-off between risk and return for portfolios that achieve the highest expected return for a given level of risk. Additionally, we include the positions of the individual assets in the risk-return space, as well as the location of the optimal portfolio. The optimal portfolio lies beyond the performance of any single asset, demonstrating the power of diversification in reducing risk while enhancing returns.

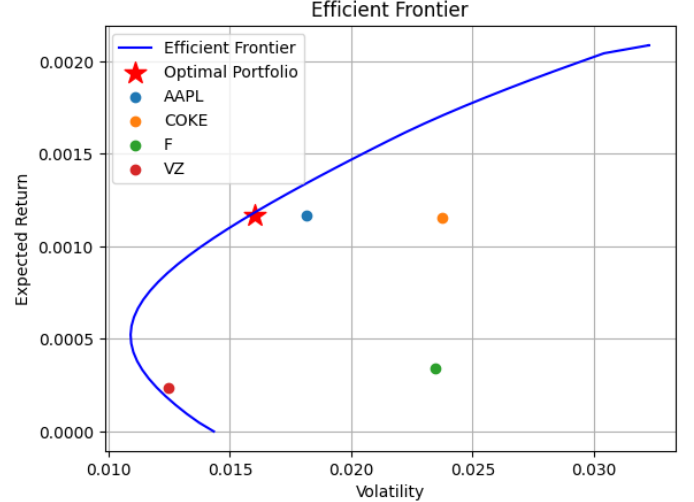


Fig. 2. Illustration of the special properties of combining assets into optimal portfolios in risk and return space

IV. SEQUENTIAL LEAST SQUARES QUADRATIC PROGRAMMING

Another approach for finding the optimal fractional share weights is Sequential Least Squares Quadratic Programming (SLSQP), implemented using the `scipy.optimize` library. Notably, this is the only method in our research that does not

rely on the CVXPY Python library. SLSQP operates by iteratively solving a sequence of quadratic programming problems, a process that introduces a degree of non-determinism into the results.

One distinct advantage of the SLSQP method is that it optimizes the Sharpe ratio directly, rather than using the return-minus-loss-aversion-adjusted-risk objective function discussed earlier. This focus on maximizing the Sharpe ratio aligns closely with the goals of many investors. However, the non-deterministic nature of SLSQP results in a wide distribution of optimal weights across runs. A single sample of this effect is documented in Table II where we run a single optimization using both QP and SLSQP for our initial dataset. This inconsistency undermines its reliability and makes it less suitable for the precise and stable weight calculations required for our research.

TABLE II
OPTIMAL WEIGHTS FOR EACH STOCK BY METHOD

Method	AAPL	F	KO	VZ
QP	0.5096	-0.1971	0.2377	0.4497
SLSQP	0.5089	-0.1962	0.2381	0.4492

V. MIXED INTEGER QUADRATIC PROGRAMMING

As an initial approach to addressing the whole share constraint, we adapted the QP procedure by switching to a mixed-integer quadratic programming (MIQP) solver and introducing a new constraint to ensure that the portfolio value does not exceed a predetermined budget. The MIQP problem was solved using the Gurobi Solver, which provided the optimal whole share portfolio weights. While Gurobi's algorithm is proprietary, it is widely believed to rely on a branch-and-bound framework to efficiently explore feasible solutions.

Initial tests revealed that the MIQP method successfully generates whole share weights corresponding to the fractional weights obtained from the QP approach for a specified γ parameter. However, it lacks a crucial feature of the QP method: the ability to scale weights to unity. This limitation means that while the QP method can adjust its solutions to ensure full investment, the MIQP method is restricted to working with a fixed level of γ . Consequently, the MIQP approach is effective for producing solutions at specific levels of risk aversion but cannot be used to identify a global optimum across the risk-return spectrum.

VI. TWO-STAGE MIXED INTEGER QUADRATIC PROGRAMMING

A. Procedure

Our proposed two-stage optimization method minimizes the deviation between the fractional weights of the optimal portfolio and the adjusted whole share weights. In the first stage, we solve the quadratic programming problem to calculate the fractional share weights. These weights are scaled to ensure

that their sum equals one, creating a fully invested portfolio. This fractional solution represents the mean-variance optimal portfolio in the absence of whole share constraints.

In the second stage, we reframe the problem as a mixed-integer quadratic programming problem. The objective in this stage is to minimize the sum of squared differences between the fractional share weights obtained in the first stage and the adjusted whole share weights, as expressed in Equation 3. Additionally, a budget constraint ensures that the total portfolio value does not exceed the specified financial limit. This two-stage framework allows us to balance theoretical optimization with practical constraints, producing a portfolio that closely approximates the ideal fractional solution while adhering to whole share requirements.

A key advantage of this approach is that the final output weights remain close to unity for any given set of asset data. This is because the first stage identifies the optimal fractional share weights, scaled to sum to one, which then serves as the target for the second stage.

$$\min_{\mathbf{w}} \sum_i (w_i^* - w_i)^2 \quad (3)$$

VII. EXPERIMENTS

Instead of relying on rigorous mathematical proofs to validate the proposed method, we demonstrate its strengths and limitations through a series of three experiments. These experiments provide practical insights into the performance of our two-stage whole share constrained optimization approach.

A. Backlog Risk

To evaluate the performance of the two-stage optimization, we introduce the concept of backlog risk. Backlog risk quantifies the additional risk introduced by discrepancies between fractional and whole share weights. It is calculated using the following formula:

$$\text{Backlog Risk} = \sqrt{(w_{\text{frac}} - w_{\text{whole}})^T V (w_{\text{frac}} - w_{\text{whole}})} \quad (4)$$

Here, w_{frac} represents the fractional weights, w_{whole} the whole share weights, and V the covariance matrix. This measure intuitively captures the excess risk stemming from deviations in the intended risk allocation. Backlog risk is widely used to assess the impact of execution and liquidity challenges in portfolio management.

B. Data

In each experiment, stock price distributions are randomly sampled with replacement from the CRSP historical daily stock price dataset. To maintain consistency and isolate the effects of whole share constraints, the covariance matrix and the vector of expected returns are generated randomly but held constant across all experiments. This approach ensures that additional randomness does not obscure the results, allowing for a clearer evaluation of the method's performance.

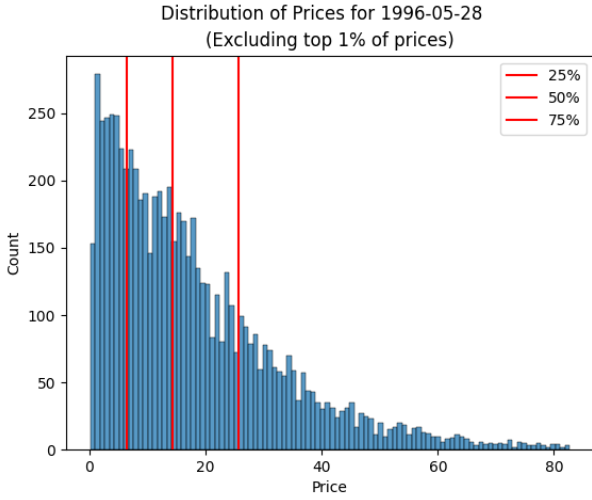


Fig. 3. A single sample of prices from a random date in the historical data set.

C. Experiment 1: Rounding Methods

The results for this experiment can be seen in Figures 4 and 5. This experiment compares the distribution of backlog risk for four portfolio construction methods: rounding up (ceil), rounding down (floor), rounding from the midpoint (mid), and our proposed two-stage method. Note that Figure 5 is a kernel density estimate plot, and so the x-axis loses its interpretability. The results show that the two-stage method achieves the narrowest and most concentrated distribution around zero, indicating that it introduces the least backlog risk. This is a significant improvement over the other methods. The rounding up and rounding down methods exhibit much wider distributions, reflecting the higher backlog risks caused by naive rounding of weights. The midpoint rounding method performs better than the round up and round down approaches with a mean backlog much closer to the two-stage method. Overall, this experiment highlights the effectiveness of the two-stage MIQP approach in reducing backlog risk, outperforming traditional rounding methods.

D. Experiment 2: Cardinality Impact

In this experiment, backlog risk is measured as the number of assets in the portfolio increases. The results reveal that backlog risk grows significantly for the rounding up and rounding down methods, particularly as the number of assets exceeds 50. This behavior underscores the poor scalability of naive rounding techniques, which introduce larger deviations as portfolio cardinality increases. The midpoint rounding method consistently exhibits lower backlog risk compared to the round up and round down methods, but it still shows notable growth as the number of assets scales up. In contrast, the two-stage MIQP method maintains the lowest backlog risk across all portfolio sizes. Even as the number of assets grows, the two-stage method MIQP remains robust and continues to produce whole share portfolios that closely approximate the fractional optimal solution. This result demonstrates the

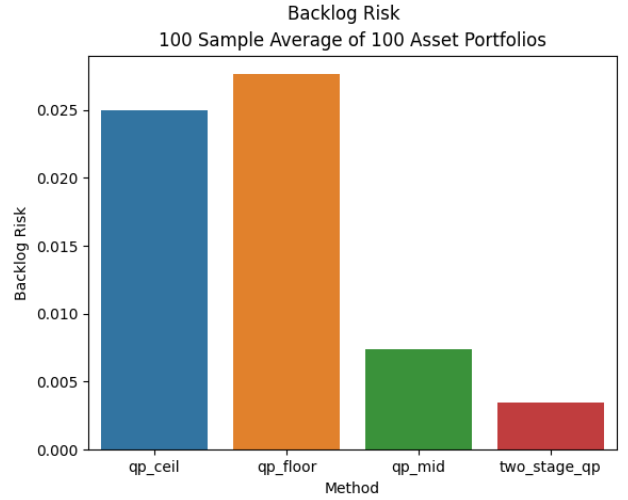


Fig. 4. Comparison of average backlog risk across different whole share optimization methods: rounding up, rounding down, rounding up and down, and two-stage MIQP.

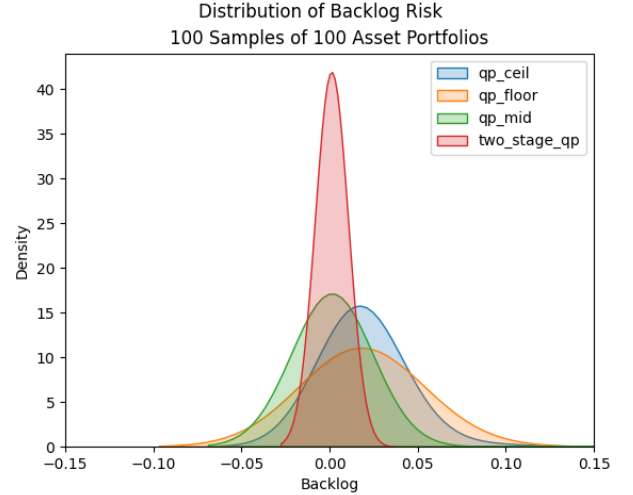


Fig. 5. Comparison of the kernel density estimated distributions across different whole share optimization methods: rounding up, rounding down, rounding up and down, and two-stage MIQP.

superior scalability of the two-stage MIQP approach, which performs reliably even for large portfolios.

However, in running these experiments, we found that the Gurobi solver struggled to find the optimal portfolio when the number of assets exceeded 100. Further research could explore more customized MIQP algorithms that could allow for higher cardinality in whole share portfolio optimization

E. Experiment 3: Budget Impact

This experiment investigates how the backlog ratio behaves as the portfolio budget increases. The results show a clear trend: as the budget scales from 100 to 1,000,000, the log backlog ratio decreases significantly. For smaller budgets, backlog risk remains relatively high, as whole share constraints impose larger deviations in the weight allocations.

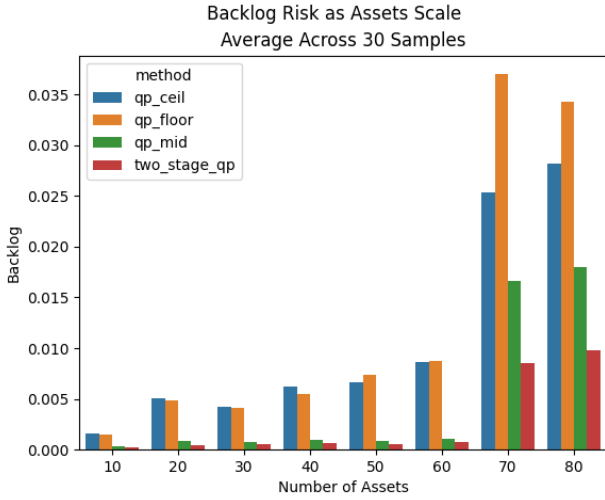


Fig. 6. The impact of scaling up the number of assets in a portfolio on its backlog

However, as the budget increases, these deviations become less significant, and the two-stage MIQP method produces near-optimal portfolios with minimal backlog risk. At very large budgets, such as 1,000,000, the backlog ratio approaches zero, indicating near-perfect alignment between the whole share portfolio and the fractional optimal portfolio. These findings illustrate that our original claim regarding large hedge funds is true. As a portfolio's budget increases, the granularity problem diminishes rapidly.

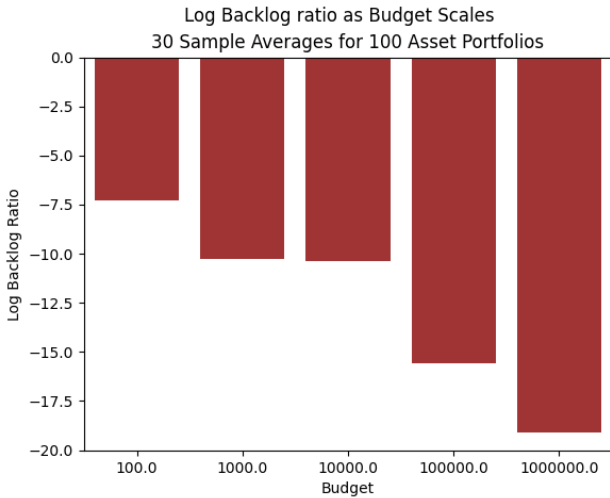


Fig. 7. Impact of scaling budget on the backlog of two-stage optimized portfolios.

VIII. CONCLUSION

This paper introduces a novel approach to mean-variance portfolio optimization that incorporates whole share constraints using a two-stage mixed-integer quadratic programming framework. By addressing the practical limitation of

fractional shares, our method offers a significant improvement over traditional rounding techniques, such as rounding up, rounding down, and midpoint rounding. Through a series of experiments, we demonstrated that the two-stage MIQP method consistently outperforms naive approaches in minimizing backlog risk and closely approximates the fractional optimal solution.

The first experiment highlights the superior precision of the two-stage method, as it achieves the narrowest backlog risk distribution compared to traditional rounding methods. In the second experiment, we showed that the two-stage method scales effectively with increasing portfolio cardinality, maintaining low backlog risk even as the number of assets grows. Finally, the third experiment reveals that backlog risk diminishes as portfolio budgets increase, validating our hypothesis that larger capital reserves mitigate the granularity problem inherent in whole share trading.

In summary, the two-stage MIQP method demonstrates its practical value for investors operating under whole share constraints. Future research could focus on improving solver efficiency for high-cardinality portfolios and exploring additional real-world constraints, such as transaction costs or liquidity considerations, to further refine the applicability of the proposed method.

ACKNOWLEDGMENT

I would like to express thanks to Dr. Grimsman, whose class on convex optimization provided the foundation for this work. I am also thankful for my finance professors, Dr. Bates and Dr. Boyer, for their invaluable insights into portfolio theory. The mentorship of these professors has been instrumental in shaping this research and my understanding of the intersection between finance and computer science.

REFERENCES

- [1] H. Markowitz, "Portfolio Selection," *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, March 1952.
- [2] W. F. Sharpe, "Mutual Fund Performance," *The Journal of Business*, vol. 39, no. 1, pp. 119–138, January 1966.
- [3] Center for Research in Security Prices (CRSP), "CRSP US Stock Database," University of Chicago Booth School of Business. Available: <https://www.crsp.org/products/research-products/crsp-stock-database>.